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10 CONCEPTUAL Q&A CLOUD · PRACTICE WORKSHEET

CONCEPTUAL SERIES

Cloud SQL

Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

CLOUD SQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Cloud

TOTAL: 10 QUESTIONS

Q1 What is Cloud SQL and what is its primary purpose in its cloud ecosystem? *Threat Model*

Q2 How does IAM work with Cloud SQL? *Architecture*

Q3 How do you monitor Cloud SQL in production? *Configuration*

Q4 How do you scale Cloud SQL to handle variable load? *Hardening*

Q5 How do you secure Cloud SQL in production? *SSO / IdP*

Q6 How do you manage configuration and secrets for Cloud SQL? *Observability*

Q7 What are the main cost drivers for Cloud SQL and how do you optimize them? *Lifecycle*

Q8 How do you implement high availability with Cloud SQL? *Compliance*

Q9 How does Cloud SQL integrate with a CI/CD pipeline? *Testing*

Q10 What are common pitfalls when first adopting Cloud SQL? *Tuning*

SOLUTIONS GUIDE
Cloud SQL

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Cloud SQL is a managed cloud
Mitigation

Cloud SQL is a managed cloud service that handles a specific infrastructure concern (compute, storage, messaging, AI). Using it shifts operational burden to the cloud provider while you focus on application logic.

A2 Access to Cloud SQL is controlled
Primitives

Access to Cloud SQL is controlled via IAM roles and policies. Attach the minimum required permissions to a service account or role. Avoid long-lived access keys; use instance roles or Workload Identity Federation instead.

A3 Cloud SQL emits metrics to the
Hardening

Cloud SQL emits metrics to the cloud provider's monitoring service (CloudWatch, Cloud Monitoring, Azure Monitor). Set alerts on key metrics (error rate, latency, capacity). Use distributed tracing to correlate across services.

A4 Cloud SQL supports auto-scaling policies based
Gotchas

Cloud SQL supports auto-scaling policies based on CPU, queue depth, or custom metrics. Set minimum and maximum capacity. Horizontal scaling (more instances) is generally preferred over vertical for cloud workloads.

A5 Place Cloud SQL in a private
Federation

Place Cloud SQL in a private subnet with no public endpoint where possible. Use VPC security groups or firewall rules to allow only required traffic. Encrypt data at rest and in transit. Rotate credentials regularly.

A6 Store sensitive values in a managed
Observability

Store sensitive values in a managed secret store (AWS Secrets Manager, GCP Secret Manager, Azure Key Vault). Inject secrets as environment variables at runtime, never bake them into container images or source code.

A7 Cost in Cloud SQL typically comes
Automation

Cost in Cloud SQL typically comes from compute hours, data transfer, storage, and request counts. Right-size instances, use reserved capacity for steady-state workloads, and enable lifecycle policies to expire old data.

A8 Deploy Cloud SQL across multiple availability
Governance

Deploy Cloud SQL across multiple availability zones. Use managed replicas or active-active configurations. Implement health checks and automatic failover. Test resilience regularly with chaos engineering or failover drills.

A9 CI/CD pipelines use the cloud CLI
Assessment

CI/CD pipelines use the cloud CLI or SDK to deploy updates to Cloud SQL after tests pass. Infrastructure-as-code (Terraform, CDK) defines Cloud SQL configuration as version-controlled code deployed via the pipeline.

A10 Common mistakes include misconfigured IAM
Optimization

Common mistakes include misconfigured IAM policies (too permissive), no cost alerting, deploying only in one availability zone, and missing backups. Read the service SLA carefully to understand what the cloud provider guarantees.